

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop **Code:**

```
#include <stdio.h>

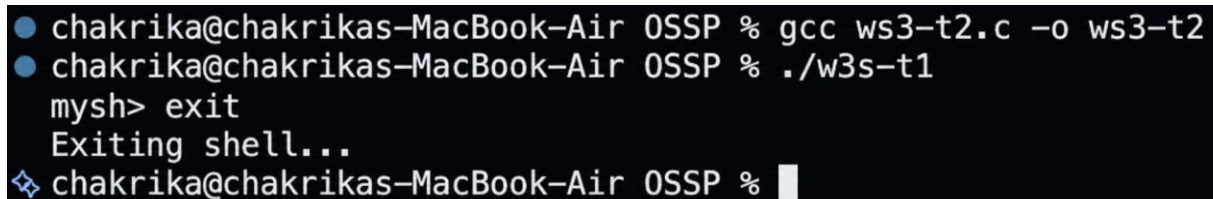
#include <string.h> #define MAX 256 int main() {
char input[MAX]; while (1) { printf("mysh>
"); fflush(stdout); if (fgets(input,
sizeof(input), stdin) == NULL) {
printf("\n"); break;
}
input[strcspn(input, "\n")] = '\0';

if (strlen(input) == 0)
continue;

if (strcmp(input, "exit") == 0) {
printf("Exiting shell...\n"); break;
}

printf("You entered: %s\n", input);
}
return 0;
}
```

Output:

A terminal window with a black background and white text. The prompt is 'chakrika@chakrikas-MacBook-Air OSSP %'. The user enters 'gcc ws3-t2.c -o ws3-t2'. The prompt is 'chakrika@chakrikas-MacBook-Air OSSP %'. The user enters './ws3-t1'. The prompt is 'mysh>'. The user enters 'exit'. The output is 'Exiting shell...'. The prompt is 'chakrika@chakrikas-MacBook-Air OSSP %' followed by a cursor.

```
● chakrika@chakrikas-MacBook-Air OSSP % gcc ws3-t2.c -o ws3-t2
● chakrika@chakrikas-MacBook-Air OSSP % ./ws3-t1
mysh> exit
Exiting shell...
❖ chakrika@chakrikas-MacBook-Air OSSP %
```

Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

1. **Open()**
2. **Read()**
3. **Write()**
4. **Close() open()**

syntax: open(" filename.txt", modes, permissions)

Example: open("First_Prgm.txt", O_CREAT | O_WRONLY, 0644);

Table 1: Modes of Open system call

Flag	Purpose
O_RDONLY	Open for read only
O_WRONLY	Open for write only
O_RDWR	Open for both read and write
O_CREAT	Create file if it does not exist
O_TRUNC	Truncate (empty) an existing file
O_APPEND	Append data to the end of the file
O_EXCL	Fail if file already exists (used with O_CREAT)

read()

Syntax: read(int fd, void *buf, size_t count);

Example: read(fd, buffer, 100)

#include <stdio.h>

```
#include <unistd.h>
int main() {   char
buffer[100];   int
n;
```

```

    printf("Enter some text: ");    n = read(0,
buffer, sizeof(buffer) - 1);    buffer[n] = '\0';
// Null-terminate the string    printf("You
entered: %s", buffer);
    return 0; }

```

write()

Syntax: write(int fd, const void *buf, size_t count);

Example: write(fd, message, 11);

```
#include <unistd.h>
```

```
int main()
```

```

{
    write(1, "Hello World\n", 12);
    return 0;
} Code:

```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <fcntl.h>
```

```
#define BUF_SIZE 128
```

```
int main() {    int fd1, fd2,
```

```

n;                                char

```

```
buffer[BUF_SIZE];
```

```

    fd1 = open("File1.txt", O_RDONLY);    if (fd1
< 0) { perror("open File1.txt"); exit(1); }

```

```

    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);
if (fd2 < 0) { perror("open File2.txt"); close(fd1); exit(1); }

    while ((n = read(fd1, buffer, BUF_SIZE)) > 0) {
        if (write(fd2, buffer, n) != n) {
perror("write");        close(fd1);
close(fd2);        exit(1);
        }
    }
    if (n < 0) perror("read");

    close(fd1);
close(fd2);

    printf("File1.txt copied to File2.txt successfully.\n");
return 0;
}

```

Output:

```

❌ chakrika@Chakrikas-MacBook-Air 0SSP % ./w3s-t2
open File1.txt: No such file or directory
❖ chakrika@Chakrikas-MacBook-Air 0SSP % █

```